

Linear regression

- Problem setup
- Least-squares solutions
- Bayesian solutions

Regression problems

We consider **input/output data**

...ie, training examples

$$\mathcal{D} = \left\{ (x_n, y_n) \right\}_{n=1}^N \subset \mathbb{X} \times \mathbb{Y}$$

with input space $x \in \mathbb{X}$ and output space $y \in \mathbb{Y}$

...ie, input features and target labels

Our goal is to obtain a **regression function**

...ie, supervised learning

$$f : \mathbb{X} \mapsto \mathbb{Y}$$

such that

$$f(x_n) \approx y_n,$$

$$n = 1, \dots, N$$

• In the typical setting, our spaces are

$$\mathbb{X} \equiv \mathbb{R}^K$$

$$\mathbb{Y} \equiv \mathbb{R}$$

and our requirement reads

$$|y_n - f(x_n)| \approx 0,$$

$$n = 1, \dots, N$$

Linear regression problems

We consider data with K -dim features and 1-dim targets

$$\mathcal{D} = \left\{ (x_n, y_n) \right\}_{n=1}^N \subset \mathbb{R}^K \times \mathbb{R}$$

Our goal is to obtain a regression function

$$f : \mathbb{R}^K \mapsto \mathbb{R}, \quad |y_n - f(x_n)| \approx 0, \quad n = 1, \dots, N$$

Our problem is ill-posed

...because it fails uniqueness

- To work around non-uniqueness, in linear regression we choose a set of basis functions

$$\phi^m : \mathbb{R}^K \mapsto \mathbb{R}, \quad m = 1, \dots, M$$

and seek only solutions of the form

...our basis functions may be non-linear

$$f(x) = \sum_{m=1}^M c^m \phi^m(x)$$

for appropriate coefficients $c^{1:M} \in \mathbb{R}$

Linear regression problems

We consider **data** $\{(x_n, y_n)\}_{n=1}^N \subset \mathbb{R}^K \times \mathbb{R}$ and a set of **basis functions** $\{\phi^m\}_{m=1}^M \subset \mathbb{R}^{\mathbb{R}^K}$, our **solution** is

$$f(x) = \sum_{m=1}^M c^m \phi^m(x) = [\phi^1(x) \quad \phi^2(x) \quad \cdots \quad \phi^M(x)] \begin{pmatrix} c^1 \\ c^2 \\ \vdots \\ c^M \end{pmatrix} = \phi(x)c$$

Here, coefficients and basis are gathered in

$$c \in \mathbb{R}^M,$$

$$\phi(x) \in \mathbb{R}^{1 \times M}$$

Our requirement reads

...the **design** matrix is Φ

$$\left. \begin{matrix} f(x_1) \approx y_1 \\ f(x_2) \approx y_2 \\ \dots \\ f(x_N) \approx y_N \end{matrix} \right\} \iff \left. \begin{matrix} \phi(x_1)c \approx y_1 \\ \phi(x_2)c \approx y_2 \\ \dots \\ \phi(x_N)c \approx y_N \end{matrix} \right\} \iff \begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_N) \end{bmatrix} c \approx \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} \iff \Phi c \approx Y$$

Ordinary least-squares solution

We seek to solve a linear system of the form

$$\Phi c \approx Y \quad \Phi \in \mathbb{R}^{N \times M}, \quad c \in \mathbb{R}^M, \quad Y \in \mathbb{R}^N$$

We cast this as a **minimization problem**

$$c_* = \operatorname{argmin}_c \|\Phi c - Y\| = \operatorname{argmin}_c \|\Phi c - Y\|^2$$

and obtain its solution via the **normal equations**

- Intuitively, this goes like this

$$\begin{aligned}\Phi c &= Y \\ \Phi^t \Phi c &= \Phi^t Y \\ (\Phi^t \Phi)^{-1} (\Phi^t \Phi) c &= (\Phi^t \Phi)^{-1} \Phi^t Y \\ c &= (\Phi^t \Phi)^{-1} \Phi^t Y \\ c &= \Phi^\dagger Y\end{aligned}$$

- Formally, this goes like this

$$\begin{aligned}0 &= \nabla_c \|\Phi c - Y\|^2 \\ &= \nabla_c (\Phi c - Y)^t (\Phi c - Y) \\ &= \nabla_c [c^t \Phi^t \Phi c - 2c^t \Phi^t Y + Y^t Y] \\ &= 2\Phi^t \Phi c - 2\Phi^t Y \\ \Phi^t \Phi c &= \Phi^t Y\end{aligned}$$

Here, $\cdot^\dagger$ denotes the Moore-Penrose **pseudo-inverse**

...the **Gram** matrix is $\Phi^t \Phi$

Ridge least-squares solution

We seek to solve a linear system of the form

$$\Phi c \approx Y \qquad \Phi \in \mathbb{R}^{N \times M}, \qquad c \in \mathbb{R}^M, \qquad Y \in \mathbb{R}^N$$

We cast this as a **minimization problem**

...the regularizer is $\lambda > 0$

$$c_* = \operatorname{argmin}_c \left(\|\Phi c - Y\|^2 + \lambda \|c\|^2 \right)$$

and obtain its solution via the **ridge normal equations**

This goes like this

$$\begin{aligned} 0 &= \nabla_c \left[\|\Phi c - Y\|^2 + \lambda \|c\|^2 \right] \\ &= \nabla_c \left[(\Phi c - Y)^t (\Phi c - Y) + \lambda c^t c \right] \\ &= \nabla_c \left[c^t \Phi^t \Phi c - 2c^t \Phi^t Y + Y^t Y + \lambda c^t c \right] \\ &= 2\Phi^t \Phi c - 2\Phi^t Y + 2\lambda c \end{aligned}$$

$$\begin{aligned} (\Phi^t \Phi + \lambda I) c &= \Phi^t Y \\ c &= (\Phi^t \Phi + \lambda I)^{-1} \Phi^t Y \end{aligned}$$

Bayesian regression

We consider **data** $\left\{ (x_n, y_n) \right\}_{n=1}^N \subset \mathbb{R}^K \times \mathbb{R}$, a **function** $\phi : \mathbb{R}^K \mapsto \mathbb{R}^{1 \times M}$, and seek **solutions**

$$f(x) = \phi(x)c$$

This is a **parameter estimation problem**

- A **basic** Bayesian model consists of

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ y_n | c &\sim \text{Normal}(\phi(x_n)c, \eta), \end{aligned} \quad n = 1, \dots, N$$

- An **extended** model consists of

$$\begin{aligned} \tau &\sim \text{Gamma}(\alpha, \psi) \\ c &\sim \text{Normal}_M(\mu, C) \\ x_n &\sim \mathbb{F}, \\ y_n | c, \tau, x_n &\sim \text{Normal}(\phi(x_n)c, 1/\tau), \end{aligned} \quad \begin{aligned} & \\ & \\ n = 1, \dots, N \\ n = 1, \dots, N \end{aligned}$$

Bayesian solution

Our **basic** Bayesian model is

$$\begin{aligned}c &\sim \text{Normal}_M(\mu, C) \\ y_n|c &\sim \text{Normal}(\phi(x_n)c, \eta), \quad n = 1, \dots, N\end{aligned}$$

This is equivalent to

$$\begin{aligned}c &\sim \text{Normal}_M(\mu, C) \\ \left(\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_N \end{array} \right) &\left| \begin{array}{c} c \sim \text{Normal}_N \left(\begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_N) \end{bmatrix} c, \begin{bmatrix} \eta & 0 & \cdots & 0 \\ 0 & \eta & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \eta \end{bmatrix} \right) \end{array} \right.\end{aligned}$$

which, in turn, takes the form

$$\begin{aligned}c &\sim \text{Normal}_M(\mu, C) \\ Y|c &\sim \text{Normal}_N(\Phi c, \eta I)\end{aligned}$$

Bayesian solution

Our **basic** Bayesian model takes the form

$$c \sim \text{Normal}_M(\mu, C)$$
$$Y|c \sim \text{Normal}_N(\Phi c, \eta I)$$

Because of **conjugacy**, our model's **posterior** is given by

$$c|Y \sim \text{Normal}_M(\mu', C')$$

and the updated **hyper-parameters** are

$$C' = \left(C^{-1} + \frac{1}{\eta} \Phi^t \Phi \right)^{-1}$$
$$\mu' = C' \left(C^{-1} \mu + \frac{1}{\eta} \Phi^t Y \right)$$

Our **MAP estimate** is

$$c_* = \mu' = \left(C^{-1} + \frac{1}{\eta} \Phi^t \Phi \right)^{-1} \left(C^{-1} \mu + \frac{1}{\eta} \Phi^t Y \right)$$

Bayesian solution

Our **basic** Bayesian model takes the form

$$\begin{aligned}c &\sim \text{Normal}_M(\mu, C) \\ Y|c &\sim \text{Normal}_N(\Phi c, \eta I)\end{aligned}$$

Our **MAP estimate** is

$$c_* = \left(C^{-1} + \frac{1}{\eta} \Phi^t \Phi \right)^{-1} \left(C^{-1} \mu + \frac{1}{\eta} \Phi^t Y \right)$$

In the **special case** of $\eta = 1$ and $\mu = 0$ and $C = \frac{1}{\lambda} I$

...this case recovers **ridge regression**

- our model is

$$\begin{aligned}c &\sim \text{Normal}_M(0, I/\lambda) \\ Y|c &\sim \text{Normal}_N(\Phi c, I)\end{aligned}$$

- our MAP estimate is

$$c_* = (\lambda I + \Phi^t \Phi)^{-1} \Phi^t Y$$